Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	A rational discovery/development platform for human microbiome-originated, precision live bacterial products (pLBPs)			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1 (**Large Category**)**		**Technology Field 2 (**Large Category**)**	
	Bio-Medical		Bio-Medical	
	Medium Category	Small Category	Medium Category	Small Category
	Medicinal Bio	Microorganisms and Enzyme Catalysts	Convergence Bio	Other Convergence Bio-products/Technologies
Concept of Proposed Technology	The proposed technology centers on a high-precision Discovery Metabolomics platform designed to decode the chemical communication between the gut microbiome and the host system. Unlike traditional metagenomic approaches that focus on bacterial identity, this platform identifies and characterizes novel, bioactive microbial metabolites. Firstly, by integrating untargeted metabolic profiling with structural elucidation, the technology transforms complex symbiotic interactions into a chemically defined "small molecule" leads. These leads are direct effector molecules produced by the interaction among microbiome, host, and the environment, rendering a positive or negative impact on a wide range of physiology and pathology. Secondly, leveraging microbiological tools and a gnotobiotic animal system, this cross-disciplinary platform can define bacterial species (either single strain or as a community) that can enhance the production of beneficial metabolites or suppress detrimental metabolites in the system.			

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Rationale for Joint R&D Project

The clinical efficacy of traditional probiotics is often inconsistent due to a lack of defined molecular mechanisms. This project aims to transition from "generic" probiotics to **"Precision Live Bacterial Products (pLBP)"** by leveraging a **Discovery Metabolomics** and Gnotobiotic Study platforms to identify the specific effector molecules that drive host health.

Molecular-centric approaches in microbiome research

Identification and structural characterization of novel metabolites produced by specific microbial species and investigation of their functions have become our fundamental means for understanding the molecular mechanisms underlying these processes. Nonetheless, because human microbiota is highly complex, approaches targeted at predetermined species may not properly reflect how microbiota process metabolites in the host niche at a systemic level. Furthermore, a large portion of the microbiome metagenome remains unannotated; Insufficient understanding of the microbial community in the target niche remains among the major limitations of the current microbiome-based therapeutic approaches.

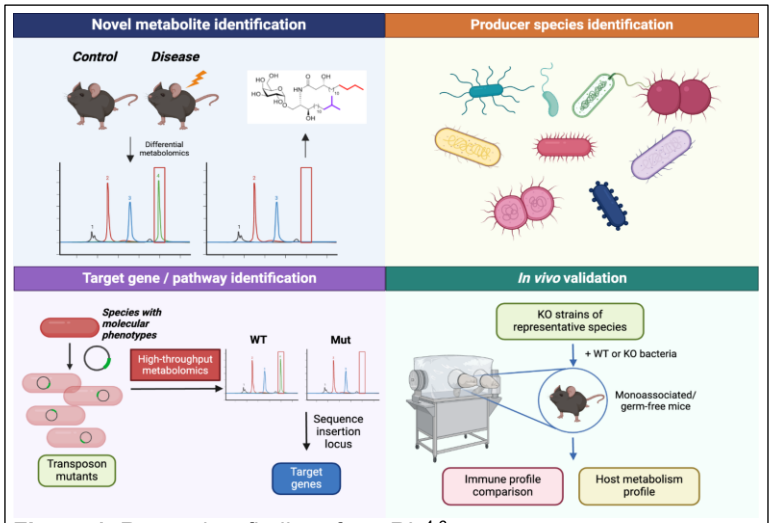

Figure 1. Recent key findings from RI. [4,6]

Mass General Brigham (Research Institution, RI) and the Microbiome metabolomics lab (PI: Sungwhan F. Oh, Ph.D.) have demonstrated strong track records in elucidating the biological functions of gut microbiota-derived metabolites. Oh lab has developed cross-disciplinary platforms that can comprehensively investigate the metagenomic and biochemical landscape of gut microbiota to address this issue (**Fig. 1**). In particular, RI established a mass spectrometry-based platform to obtain quantitative metabolomic profiles of human microbiota. RI also investigated biochemical mechanisms used by the candidate metabolite-producing species and elucidated previously unannotated pathways with a forward genetics-based approach. We have further identified and characterized microbiome-derived molecules and synthesized authentic molecules with total organic synthesis. Using these molecules with variable structures designed by nature, we have conducted multiple investigations at *in vitro* and *in vivo* levels, using synthetic molecules or gnotobiotic strains with or without

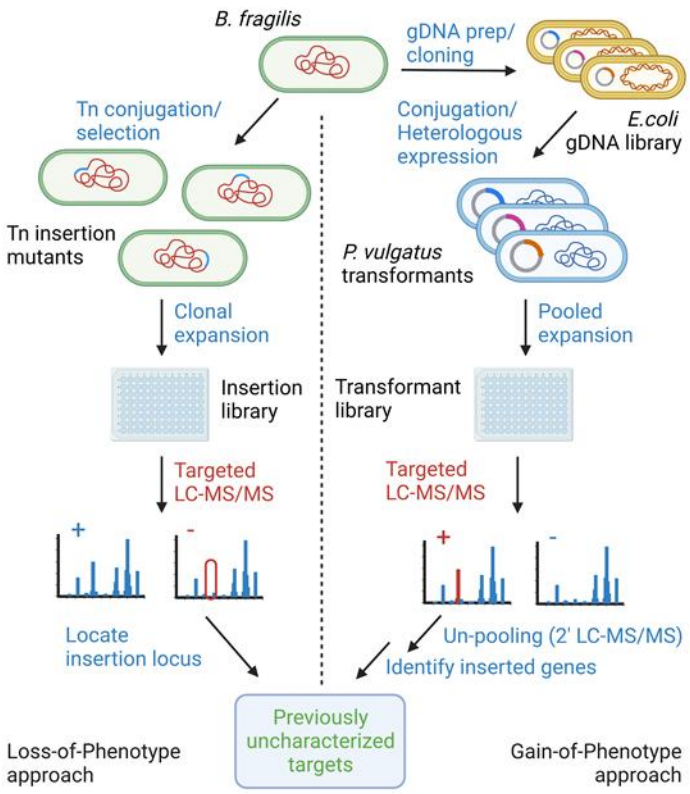

Figure 2. A scheme of parallel, forward-genetic approaches using high-throughput metabolic screening to identify previously unannotated bacterial gene (*B. fragilis agcT*).

biochemical activities of interest.

Chemical and genetic characterization of unannotated metabolites of commensal microbial origins

We have previously reported that alpha-galactosylceramides of ubiquitous human gut symbiont *B. fragilis* (BfaGCs) modulate host colonic immune development by suppressing natural killer T cell proliferation in early life[1]. Although sphingolipid biosynthesis in gut symbionts and their host impact has been rigorously addressed recently[2,3], microbial species-specific enzymatic pathways have been only partially elucidated. We have utilized forward genetic approaches (screening large number of mutants to identify target molecular phenotype) (Fig 2), to characterize *B. fragilis*-specific alpha-galactosyltransferase (agcT), along with key enzymes of sphingolipid synthesis by reverse genetic approaches (homology-based gene identification).

Structure-activity relationship (SAR) study of BfaGC and host dietary factor contribution to bioactivity.[4]

B. fragilis produces multiple structural variants (chain length and terminal branching). Using chemically defined bacterial metabolites, either biogenically purified or synthetically prepared, structure-activity relationship (SAR) studies were carried out to identify unique functional moieties (branched sphinganine group, shown in red in Fig 3). Branched-chain fatty acids are downstream products of branched-chain amino acids (BCAA) converted by BCAA acyltransferase (BCAT)[5]. Of note, BfaGC structures can be dictated by availability of exogenous BCAA, which is also confirmed in vivo by inclusion of isotope-labeled BCAA (d3-Leucine) to BfaGCs produced by *B. fragilis* in monocolonized animals (Fig 4). Genetic deletion of *B. fragilis* BCAT ortholog (BF9343-Δ3671) produced less branched BfaGC *in vitro* and *in vivo*(Fig. 5). Furthermore, BCAT KO monocolonized mice have elevated NKT cell levels, implying NKT regulation impairment with inactive BfaGCs.

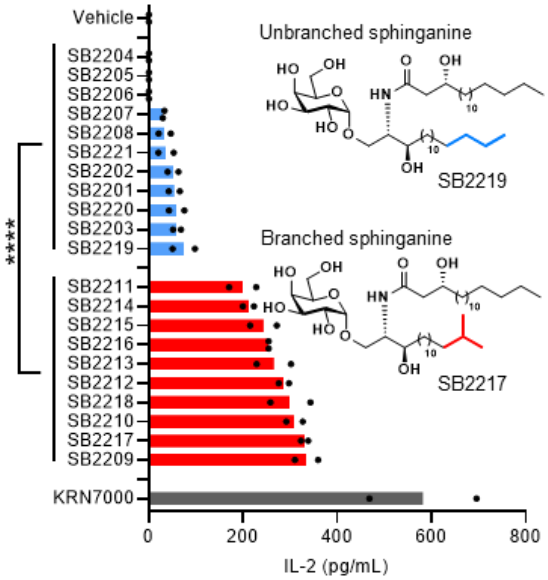

Figure 3. APC-NKT coculture assays of 21 synthetic BfaGC show that sphinganine-branched homologs (e.g., SB2217, shown as red bars) are active inducer of IL-2, where unbranched sphinganine BfaGCs (e.g. SB2219, shown as blue bars) are not. Results represent three independent sets. Throughout proposal, *; p< 0.05, **; p<0.01, ***; p<0.001 and ****; p<0.0001.

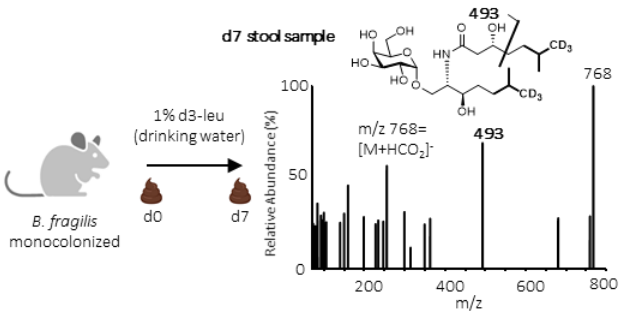

Figure 4. Host dietary BCAA is directly incorporated into BfaGCs produced by *B. fragilis* in the large intestine. The MS/MS fingerprint (m/z = 493) of d6-C34 BfaGC confirms d3-leucine incorporation into both sphinganine and fatty acyl chains. MS spectrum represents n=4.

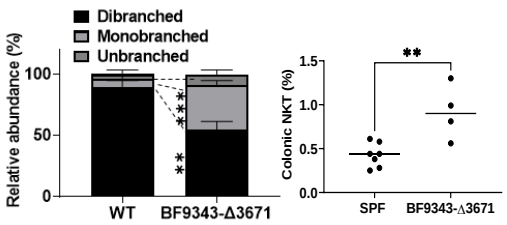

Figure 5. (Left) BF9343-Δ3671 monocolonization exhibits less branched BfaGCs (N=4). (Right) BF9343-Δ3671 monocolonization at birth shows dysregulated colonic NKT cell levels.

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	These results show that the gut symbionts utilize host dietary factors to produce bioactive primary metabolites, which feedback to host physiology. RI's ability to dissect molecular mechanisms, chracterize metabolites in complex matrices and validate their physiological impact using gnotobiotic models significantly de-risks the discovery phase of the pLBP pipeline. RI has experience and expertise in the structural characterization of novel metabolites and its mechanistic proof-of-concept by gnotobiotic validation (TRL 4). To synergize, it is expected that the Korean SME partner (KSP) provides multiple human symbiotic candidate strains with beneficial impact in human pathology, to identify and characterize keystone metabolites. In parallel, KSP also can bring in a large-scale microbial strain bank that can serve as a library for targeted metabolomic screening, to identify strains producing beneficial or metabolizing harmful metabolites of interest. Following cross-disciplinary collaboration with RI to identify key metabolites and elucidate their molecular mechanism, KSP will scale up production of specific candidate strains and optimize formulation of pLBP, targeting expanded preclinical studies (TRL 6).
Key Words	1. Symbiont-Derived Small Molecules 2. Discovery Metabolomics 3. Mechanism-based, Microbiome-originated Drug Discovery 4. Human Pathophysiology 5. Molecular Translation

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (6)

R&D Objectives

1. **Establishment of a Discovery Metabolomics Pipeline**
   - Deploy untargeted LC-MS/MS workflows to profile metabolites (both annotated and previously unknown) from the candidate strains prioritized from the KSP strain bank that correlate with beneficial host physiological responses across diverse pathologies.

2. **Structural Elucidation and Molecular Characterization**
   - Identify, isolate, and characterize structures of microbial metabolites. Determine the precise chemical structures of lead metabolites, focusing on novel lipid mediators and small-molecule effectors.
   - Define the **Structure-Activity Relationship (SAR)** to understand how specific molecular variations impact therapeutic potency.

3. **Mechanistic Validation via Gnotobiotic Systems**
   - Conduct "Proof-of-Mechanism" (PoM) studies using germ-free and gnotobiotic animal models to confirm that identified metabolites are the direct drivers of host phenotypes.
   - Confirm bioactivity at strain level, to select target strain.

4. **Optimization of Precision Microbial Production**
   - Identify optimal conditions and metabolic precursors required to maximize the production of target effectors within the **pLBP**.
   - Conduct functional assays to verify that metabolic activity is maintained during the transition from laboratory to pilot-scale production.

5. **Advanced Formulation and pLBP Prototype Development**
   - Develop stabilization and encapsulation technologies to protect the viability and metabolic machinery of the **pLBP** candidates.
   - Standardize the final **pLBP** prototype for expanded preclinical validation, ensuring consistent delivery of the "molecular payload" to target physiological sites.

6. **Preclinical Validation (Target TRL 6)**
   - Execute statistically significant preclinical study in relevant disease models to establish safety, dose-response, and efficacy profiles.
   - Compile a comprehensive data package to support subsequent IND (Investigational New Drug) applications.

R&D Target Specifications

1. **RI-Led: pLBP Discovery Platform**
   - **Discovery Metabolomics Workflow:**
     - **Instrumentation:** High-Resolution Liquid Chromatography-Mass Spectrometry (HR-LC-MS/MS) for non-targeted "orphan" metabolite identification.
     - **Molecular networking :** Application of bioinformatic pipelines to annotate ipids/metabolites.
     - **Structural Elucidation:** Determination of exact structures effector molecules—critical for ensuring the "precision" of the final pLBP.
   - **Gnotobiotic Validation System:**
     - **Environment:** Sterile, Germ-Free (GF) isolator systems allowing for the

	colonization of specific bacterial consortia (Mono- or Oligo-colonization). ○ **Dietary Control:** Precision-defined diets to isolate the interaction between specific dietary precursors and microbial metabolic output. ○ **Pathological Mapping:** High-dimensional immune/metabolic profiling (e.g., flow cytometry, RNA-seq) to confirm the metabolite-host receptor interaction. **2. <u>KSP-Led: Industrial Translation & Scaling</u>** • **Functional Screening:** ○ **Assay Platforms:** Development of *in vitro* and *ex vivo* reporter assays to rapidly screen the strain bank for candidates that trigger target physiological pathways. • **Scale-up:** ○ **Process Optimization:** Identification of "Metabolic Inducers" (media additives) that maximize the pLBP's production of the effector molecule at scale-up processing and *in vivo* setting. ○ **Biomass Standardization:** Establishment of QC parameters for microbial viability and metabolic potency at a pilot-scale (e.g., 50L–500L bioreactors). • **Advanced Formulation:** ○ **Stabilization:** Techniques designed to protect the pLBP's metabolic machinery from gastric acidity and oxygen exposure.
R&D Plan	**1. <u>Research Strategies</u>** Collaboration of RI and KSP may begin in two parallel mechanisms, by targeting potentially beneficial strain candidates (Path 1) or biologically relevant metabolites (Path 2). **Path 1. Discovery metabolomics of causative metabolites from pLBP candidate strains.** Differential metabolomic analysis of candidate strains and their taxonomically related inactive controls will be conducted. Both annotated metabolites and unannotated features with specific information (such as tandem mass spectrum data) will undergo molecular networking and other bioinformatic analysis to confirm their structures. **Path 2. Targeted metabolomic profiling of the strain library.** The strain library will be incubated in diverse nutritional conditions and oxygen levels (aerobic/anerobic/microaerophilic), to quantify metabolites of interest. Strains that significantly alter target metabolites will be prioritized. ***In vivo* validation of candidate strain and species** After the initial characterization in both path 1 and path 2, detailed information about metabolite structures and strains will be obtained. A prioritized list of candidate strains will be validated by gnotobiotic model (monocolonization of germ-free animals with candidates) to confirm the production of metabolites *in vivo*. ***In vitro* and *in vivo* mechanistic studies** By using functional assays and gnotobiotic animals, beneficial impact of metabolites and candidate strains will be validated. Further studies will expand non-gnotobiotic animals to confirm bioactivity and ensure engraftment (single strainvs mixture as a stable bacterial community) **Scale-up production of candidate strains** Production of metabolically consistent candidate strains will be ensured to meet required level to carry out later in vivo preclinical validation. **2. <u>Timeline and Milestone</u>** **I. Phase 1 (0-6mo)** - Generating a detailed collaboration plan between RI and KSP - Proof-of-principle validation of lab-strain-based studies with KSP strains - Agreement in primary lead strains and target metabolites to investigate - Material Transfer Agreement / IP Agreement **II. Phase 2, Year 1 (6-18 mo): Candidate identification and validation** **Focus:** Leveraging the "Rational" discovery platform to identify high-potential pLBP candidates.

- **12 months**
  - Perform **Discovery Metabolomics** on target strains to identify novel bioactive metabolites and conduct **Structural Elucidation** to ensure chemical novelty and biological relevance.
  - Perform **Targeted Metabolomic Screening** to identify strains producing target metabolites.
  - Develop **Functional Assays** to validate the bioactivity of RI-identified metabolites in vitro.
  - Initiate **Strain Bank Mining** to identify bacterial species with high genetic potential for targeted metabolite production.
- **18 months**
  - Confirm and optimize *in vivo* production using **gnotobiotic model**
  - Selection of 1–2 lead strains with confirmed PoM and structural data.

III. Year 2 (18-30mo): Process Development & Preclinical Expansion

Focus: Transitioning from "lab-scale" discovery to a standardized, reproducible product profile.

- **24 months**
  - Complete initial gnotobiotic **Proof-of-Mechanism (PoM)** for the lead pLBP candidate.
  - Deepen mechanism studies using specialized gnotobiotic models (e.g., tissue-specific immune profiling).
  - **Scale-up Optimization:** Perform initial optimization (media screening, growth kinetics) in 5L–10L bioreactors.
- **30 months**
  - Provide structural "fingerprints" for quality control (QC) marker development.
  - **Expanded Preclinical Validation:** Test the lead pLBP in diverse disease models (beyond initial PoM) to establish a therapeutic window.

IV. Year 3 (30-36mo): Industrial Validation & Preparation for TRL 6

Focus: Demonstrating stability, scalability, and preparing for IND-enabling studies.

- **36 months**
  - Successful large-scale with consistent metabolite yield; expanded efficacy data
  - Completion of TRL 6 validation
  - Final R&D report

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• **Untargeted Metabolomics Platform:** Performing untargeted metabolic profiling on the "active" candidate strains prioritized by the KSP's functional screening. • **Targeted, high-throughput Metabolomic screening**: Performing high-throughput analysis to identify candidate strains from KSP's strain library. • **Structural Elucidation**: Utilizing high-resolution analytical tools (e.g., NMR, LC-MS/MS) to characterize the precise chemical structures of the novel metabolites identified as bioactive leads. • **Molecular Target Identification**: Investigating host-side receptors and signaling pathways triggered by the identified metabolites to complete the mechanistic picture. • **Primary *In Vivo* Validation (Gnotobiotic Systems)**: Executing precision experiments in germ-free and gnotobiotic animal models to provide the definitive "Proof-of-Mechanism" (PoM) for the effector molecules.
Expected R&R of Korean SME Partner	• **Strain Bank Provision & Mining**: Managing the large-scale library of human symbiotic bacterial strains and providing the biological material for the initial discovery phase. • **Functional Assay Development & Screening**: Designing and executing functional assays (in vitro or ex vivo) to screen the strain bank and identify candidates with high therapeutic potential. • **Bioprocess Development & Scale-up**: Translating the laboratory-scale findings into industrial processes, ensuring that the "precision" metabolic traits are preserved during mass production. • **Formulation & Stabilization**: Developing optimized delivery systems (e.g., encapsulation, lyophilization) to maintain the viability and metabolic potency of the pLBP candidates. • **Expanded Preclinical Evaluation (TRL 6)**: Executing in-depth preclinical studies in relevant disease models to establish dose-response, safety profiles, and therapeutic efficacy.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	**1. Economic Impact** • **Market Transformation:** The global human microbiome market is projected to reach **USD 0.85 billion in 2026** and surge to **USD 3.27 billion by 2035**, with a CAGR of 16%. By shifting from low-margin dietary supplements to high-value, patentable pLBPs with defined MoAs, this project allows the KSP to capture the most lucrative segment of this growth—the **Microbiome Therapeutics** sector, which is expected to witness a 32% growth rate. • **De-risking and Efficiency:** Moving the technology to **TRL 6** significantly reduces the "Valley of Death" in drug development. By identifying the exact effector molecules via RI's **Discovery Metabolomics** platform before large-scale trials, we minimize the high costs and failure rates associated with the clinical validation of traditional, uncharacterized probiotics. • **Competitive Advantage for Korean Biotech:** This collaboration enables a Korean SME to upgrade its raw "Strain Bank" into a **high-value therapeutic pipeline**. It fosters international trade and positions the KSP as a specialized leader in the "Precision Probiotics" market, which is seeing rapid growth in the Asia-Pacific region. **2. Social Impact** • **Addressing Chronic Disease Burden:** In 2026, chronic conditions like IBD, metabolic syndrome, and neuro-inflammatory disorders continue to drive a **10.9% global increase in healthcare costs**. pLBPs provide a "Precision Medicine" approach, offering personalized and sustainable solutions for these refractory pathologies. • **Reduction in Healthcare Waste:** Traditional "one-size-fits-all" probiotics often lead to inconsistent patient outcomes. By providing mechanism-defined products, this project ensures **higher treatment efficacy**, reducing the "trial-and-error" healthcare expense and the secondary costs of untreated chronic inflammation. • **Bio-inspired Safety:** As live biotherapeutics, pLBPs utilize co-evolved human symbionts, offering a safer alternative to synthetic drugs with systemic side effects. This contributes to public health by providing a new class of **"Precision Preventative"** therapies that align with the 2026 trend toward longevity and proactive wellness.

6. References

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